

UNSW Mathematics Society $\times$ Competitive
Programming and Mathematics Society Presents
MATH3711/5706 Seminar



Presented by Jeff, Brendan, Justin and Daniel

Attendance

Please scan the QR code; please let us know about any dietary requirements (for the pizza!).



1. Rings

What is a ring?

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Rings

Definition

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We say a ring is commutative if the multiplication operation is commutative. *For this course, we assume that $1 \neq 0$ in a ring.

'Igor' Rings vs 'Alex' Rings

If you've done MATH3431, you might notice that the definition of a ring differs slightly. Mainly, an 'Igor' ring (rng) does not have to include a multiplicative identity, but in this course we must have one. Why?

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The existence of a multiplicative identity is quite useful to prove properties about rings. But more on that later.

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Useful properties

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Similar to subgroups, the intersection of subrings is also a subring; this should be easy to show!

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which is a group under multiplication.

If $R^\times = R \setminus \{0\}$ then R is called a division ring, and a commutative division ring is a field. Only non-zero elements could be units, since we assume $0 \neq 1$.

Ring Homomorphisms

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Similarly, we also define

$$\begin{aligned}\text{im}(\phi) &= \{\phi(r) : r \in R\}, \\ \ker(\phi) &= \{r \in R : \phi(r) = 0\},\end{aligned}$$

and the image is a subring of S and the kernel is a subring of R .

2. Ideals

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We say R is a simple ring if the only ideals of R are $\{0\}$ and R . We can also generate ideals with certain elements, similar to generating subgroups.

Properties of Ideals

Properties

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- $I = S \iff 1_S \in I$
- $\phi^{-1}(I) \trianglelefteq R$
- $\ker(\phi) = \phi^{-1}(\{0\}) \trianglelefteq R$
- The intersection of ideals of a ring is also an ideal.

Example on Ideals

Question

Prove that every simple commutative ring is a field.

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Solution

Suppose that R is a simple commutative ring. Then $\langle r \rangle = R$ for non-zero r , so $rr' = 1$ for some $r' \in R$. Since every non-zero element generates R , every non-zero element is invertible and we are done!

Quotient Rings

Definition

If $I \trianglelefteq R$ then R/I is called the quotient ring of R by I , which is a ring with elements $r + I, r \in R$ and multiplication defined as

$$(r + I)(r' + I) = rr' + I, \quad r, r' \in R.$$

We also have the multiplicative identity $1 + I$. These 'cosets' behave similar to the cosets we observed in group theory under addition.

Isomorphism Theorems

First Isomorphism Theorem

If $\phi : R \rightarrow S$ is a ring homomorphism, then ϕ induces an isomorphism $\bar{\phi} : R / \ker(\phi) \rightarrow \text{im}(\phi)$ given by

$$\bar{\phi}(r + \ker(\phi)) = \phi(r),$$

that is, $R / \ker(\phi) \cong \text{im}(\phi)$.

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Second Isomorphism Theorem

If S is a subring of R and $I \trianglelefteq R$ then $S + I$ is a subring of R , $S \cap I \trianglelefteq S$ and we have $\frac{S}{S \cap I} \cong \frac{S + I}{I}$.

Isomorphism Theorems Continued...

Third Isomorphism Theorem

If $I, J \trianglelefteq R$ and $J \subseteq I$ then $\frac{R/J}{I/J} \cong R/I$.

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Third Isomorphism Theorem

If $I, J \trianglelefteq R$ and $J \subseteq I$ then $\frac{R/J}{I/J} \cong R/I$.

Fourth Isomorphism Theorem

If $J \trianglelefteq R$ and $\pi : R \rightarrow R/J$ is the quotient map, then π induces a bijection between the ideals of R containing J and ideals of R/J . That is, the map $I \mapsto I/J$ is bijective for $I \trianglelefteq R, J \subseteq I$.

Chinese Remainder Theorem

Theorem

Let R be a ring, and $I_1, \dots, I_n \trianglelefteq R$ such that $I_i + I_j = R$ for all $i \neq j$. If $I = \cap_{i \in [n]} I_i$ then the ring homomorphism

$$\phi : R/I \rightarrow R/I_1 \times \cdots \times R/I_n, \quad \phi(r + I) = (r + I_1, \dots, r + I_n)$$

is an isomorphism.

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Corollary

If $n_1, \dots, n_k$ are pairwise coprime positive integers and $n = n_1 \cdots n_k$, then

$$\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z} \times \cdots \times \mathbb{Z}/n_k\mathbb{Z}.$$

3. Further Ring Theory

Polynomial Rings

Definition

Let R be a ring. A polynomial p with coefficients in R is a function $p : \mathbb{N} \rightarrow R$ such that $p(n) = 0$ for all but finitely many $n \in \mathbb{N}$. We denote the set of polynomials over R as $R[x]$ and addition and multiplication is given by

$$(p_1 + p_2)(n) = p_1(n) + p_2(n),$$

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It is easy to verify that this is a ring, and to extend this definition to multiple variables $R[x_1, \dots, x_n]$. Why do we define polynomials this way?

Example on Polynomial Rings

Question

Let $\text{Aut}(R)$ denote the set of ring automorphisms on R . Find $\text{Aut}(\mathbb{Z}[x])$.

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Solution

Let $\phi \in \text{Aut}(\mathbb{Z}[x])$. Then, $\phi|_{\mathbb{Z}} = \text{id}$ since $\mathbb{Z} = \langle 1 \rangle$ and ring homomorphisms map identities to each other. Now, every non-constant polynomial is generated by x , so consider $\phi(x) = p(x)$, for some $p(x) \in \mathbb{Z}[x]$.

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$$\deg(\phi^{-1}(p(x))) = \deg(\phi^{-1}(x)) \deg(p(x)) = \deg(x) = 1.$$

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So we write $p(x) = ax + b$ for $b \in \mathbb{Z}, a \in \{1, -1\}$ and so

$$\text{Aut}(\mathbb{Z}[x]) := \{x \mapsto ax + b : b \in \mathbb{Z}, a \in \{1, -1\}\}.$$

Ring Adjunction

Definition

If R is a subring of a commutative ring S , with $a_1, \dots, a_n \in S$, then

$$R[a_1, \dots, a_n] = \text{im}(\varepsilon_{a_1, \dots, a_n} : R[x_1, x_2, \dots, x_n] \rightarrow S).$$

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Maximal Ideals

Definition

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Properties

- R/M is simple
- M is maximal if and only if R/M is a field.

Maximal Ideals and Fields

Fact

We claim that if R/M is a field, then M is maximal.

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Proof

Every field is simple, since for every non-zero element $a \in I \trianglelefteq \mathbb{F}$ implies $aa^{-1} = 1 \in I$ and so $I = \mathbb{F}$. Since R/M is a field, its only ideals are R/M and $\{M\}$. By fourth isomorphism theorem, the only ideals of R containing M are R and M , so M is maximal.

Integral Domains

Definition

If R is a commutative ring, then we call R an integral domain if $ab = 0 \implies a = 0$ or $b = 0$; no zero divisors.

Properties of Integral Domains

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- Cancellation: for non-zero $r \in R$ we have $rs = rs' \implies s = s'$ for all $s, s' \in R$.
- R is isomorphic to a subring of a field. Why?

Example on Integral Domains

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Solution

Suppose R is a finite integral domain, then it suffices to show that every non-zero element is a unit. Now, consider the map $\phi_r : R \rightarrow R, \phi(a) = ar, r \neq 0$, which is injective as

$$\phi_r(a) = \phi_r(a') \implies ar = a'r \implies a = a'$$

and so must be a bijection since the domain and codomain are both R .

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and so must be a bijection since the domain and codomain are both R . This implies $ar = 1$ for some $a \in R \setminus \{0\}$. Since r is any non-zero element, it follows that every element is a unit and we are done!

Fraction Fields

Definition

For an integral domain R , define

$$\hat{R} = R \times (R \setminus \{0\}) = \{(a, b) | a, b \in R, b \neq 0\}$$

and the relation $\sim$ by $(a, b) \sim (a', b')$ if and only if $ab' = ba'$. It is easy to verify this is an equivalence relation.

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Now, we write $K(R)$ for the set of equivalence classes of $\hat{R}$ under $\sim$ and $[a, b]$ for the equivalence class containing (a, b) . Then, we can define the operations $+, \times : K(R) \times K(R) \rightarrow K(R)$ by

$$\begin{aligned}[a, b] + [a', b'] &= [ab' + ba', bb'], \\ [a, b][a', b'] &= [aa', bb'],\end{aligned}$$

and this can be verified to be well-defined. We can also verify this is a ring, and also a field, why?

Principal Ideal Domains

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Non-examples include: $\mathbb{F}[x, y]$, $\mathbb{Z}[x]$.

Prime Ideals

Definition / Lemma

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Prime Ideals

Definition / Lemma

For an ideal of a ring R $P \trianglelefteq R$, we say P is a prime ideal if one of the following hold:

- $ab \in P \implies a \in P$ or $b \in P$,
- R/P is an integral domain.

Example on Prime Ideals

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Prove that in a finite commutative ring, every prime ideal is maximal.

Solution

Let R be a finite commutative ring, and $P \trianglelefteq R$ be a prime ideal of R . Since every finite integral domain is a field, it suffices to show that R/P has no zero divisors. So, suppose that $a + P, b + P$ are zero divisors; I.E $(a + P)(b + P) = P$. Then $ab + P = P$ and so $ab \in P$. But since P is a prime ideal, either $a \in P$ or $b \in P$, so one of our starting elements is a zero, hence R/P is an integral domain and we are done.

4. Factorisation Theory

Factorisation of Elements in R

Definition

In an integral domain R , for $r, s \in R$, we say that $r \mid s$ if $s = rt$ for some $t \in R$. If $t \in R^\times$ then r, s are called associates.

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Proposition

For r, s in an integral domain R , we have that

- $r \mid s$ if and only if $\langle s \rangle \subseteq \langle r \rangle$,
- r, s are associates if and only if $\langle s \rangle = \langle r \rangle$,

Example Putting Everything Together

Question

Find the ideals of $\mathbb{C}[x]/\langle x^n \rangle$ for $n \in \mathbb{N}$.

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Solution

Consider the ideals of $\mathbb{C}[x]$ which contain $\langle x^n \rangle$, and note that $\mathbb{C}[x]$ is a principal ideal domain, so all ideals will be of the form $\langle p(x) \rangle$. If $\langle x^n \rangle \subseteq \langle p(x) \rangle$, then $p(x) \mid x^n$. The factors of x^n are $1, x, x^2, \dots, x^n$, so the ideals of $\mathbb{C}[x]$ containing $\langle x^n \rangle$ are $\langle 1 \rangle, \langle x \rangle, \dots, \langle x^n \rangle$. By the fourth isomorphism theorem, the image of these ideals under the quotient map correspond to the ideals of $\mathbb{C}[x]/\langle x^n \rangle$, so its ideals are $\mathbb{C}[x]/\langle x^n \rangle, \langle x^{n-1} \rangle/\langle x^n \rangle, \langle x^{n-2} \rangle/\langle x^n \rangle, \dots, \langle x \rangle/\langle x^n \rangle, \{ \langle x^n \rangle \}$.

Factorisation of Elements in R

Definition

We say that a non-zero, non-unit element in R is irreducible if its only factors are its associates and units. And, we say that a non-zero, non-unit element $p \in R$ is prime if $p|rs$ implies either $p|r$ or $p|s$.

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Proposition

Suppose that $p \in R \setminus (R^\times \cup \{0\})$, then p is irreducible if and only if the only principal ideals containing $\langle p \rangle$ are $\langle p \rangle$ and R . And, p is prime if and only if $\langle p \rangle$ is a prime ideal.

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We say that a non-zero, non-unit element in R is irreducible if its only factors are its associates and units. And, we say that a non-zero, non-unit element $p \in R$ is prime if $p|rs$ implies either $p|r$ or $p|s$.

Proposition

Suppose that $p \in R \setminus (R^\times \cup \{0\})$, then p is irreducible if and only if the only principal ideals containing $\langle p \rangle$ are $\langle p \rangle$ and R . And, p is prime if and only if $\langle p \rangle$ is a prime ideal.

Lemma

Every prime element is irreducible.

5. Unique Factorisation Domains

Unique Factorisation Domains

Over the positive integers, we have a fundamental theorem of arithmetic, stating that every integer is uniquely the product of prime numbers. We want to make an analagous statement for rings. But this notion does not hold for all rings, i.e., $\mathbb{Z}[\sqrt{-5}]$ factors 6 into 3×2 or $(1 + \sqrt{-5})(1 - \sqrt{-5})$. Each of these elements are *prime* so we do not have unique factorisation.

There is a specific category of domains that have the unique factorisation property.

Unique Factorisation Domains

Factorisation Theory

Consider a commutative domain R and let $r, s \in R$.

- We say that r is a **factor** of s , or that r divides s , if there exists some $t \in R$ such that $s = rt$.
- We say that r and s are **associates** if there exists a unit $u \in R^*$ such that $r = su$.
- We say that a non-zero and non-unit element r is **irreducible** if every factor of r is either a unit or an associate of r .
- We say that a non-zero and non-unit element p is **prime** if $p \mid rs$ implies that $p \mid r$ or $p \mid s$ for all $r, s \in R$.

Unique Factorisation Domains

Properties

- $r \mid s$ if and only if $\langle s \rangle \subseteq \langle r \rangle$.
- r and s are associates if and only if $\langle r \rangle = \langle s \rangle$.
- A non-zero and non-unit element p is irreducible if and only if the only principal ideals that contain $\langle p \rangle$ are $\langle p \rangle$ and $\langle 1 \rangle$.
- A non-zero and non-unit element p is prime if and only if $\langle p \rangle$ is a prime ideal.

In a commutative domain, every prime element is irreducible. The converse, however, is not necessarily true; that is, irreducible elements are not necessarily prime.

Unique Factorisation Domains

Unique Factorisation Domains (UFDs)

Let R be a commutative domain. We say that R is a *unique factorisation domain* if the following hold:

- For any non-zero and non-unit element $r \in R$, we have that

$$r = p_1 p_2 \dots p_m$$

where p_i is irreducible for $1 \leq i \leq m$.

- If we have two factorisations of the same element $r = p_1 p_2 \dots p_m = q_1 q_2 \dots q_n$, then $m = n$ and we can re-index q_i so that q_i and p_i are associates for all i .

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Equivalent characterisation

Let R be an integral domain. R is a UFD if and only if (UD1) holds and every irreducible element is prime.

Unique Factorisation Domains

Factorisation over $\mathbb{Z}$

We see that $10 = 5 \times 2$ but we can also factor 10 as $10 = -5 \times -2$. -5 and 5 are associates, and 2 and -2 are associates.

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Show that 21 does not factor uniquely as a product of irreducibles in $\mathbb{Z}[\sqrt{-5}]$.

We have that $21 = 7 \times 3 = (4 + \sqrt{-5})(4 - \sqrt{-5})$. We now wish to show that each of the factors is irreducible.

Suppose that $7 = (a + b\sqrt{-5})(c + d\sqrt{-5})$ for some $a, b, c, d \in \mathbb{Z}$. Then we can take the complex $|\cdot|^2$ of both sides to get

$$49 = (a^2 + 5b^2)(c^2 + 5d^2).$$

Unique Factorisation Domains

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49 has divisors $\{1, 7, 49\}$ and we can exhaust cases. If $a^2 + 5b^2 = 1$, then $a = \pm 1, b = 0$, which gives a unit factor. Same for $c^2 + 5d^2$. Thus, $c^2 + 5d^2 = a^2 + 5b^2 = 7$. But we can again easily show that this has no integer solutions.

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We can extend this approach to the other three factors to show that 21 does not factor uniquely as a product of irreducibles.

Principal Ideal Domains

Principal Ideal Domains (PIDs)

Let R be a commutative domain. R is a principal ideal domain if every ideal is principal.

Properties of PIDs

- Let R be a PID. Then every irreducible element in R is prime.
- Let R be a PID. Then every non-zero and non-unit element $r \in R$ can be factored into a product of irreducible elements.

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- Let R be a PID. Then every irreducible element in R is prime.
- Let R be a PID. Then every non-zero and non-unit element $r \in R$ can be factored into a product of irreducible elements.

As seen in the lectures, the above two properties is a sufficient characterisation of UFDs, i.e., it turns out that PIDs are UFDs. So we have

$$\text{ED} \subseteq \text{PID} \subseteq \text{UFD}.$$

(Tut 8, Q3)

Show that $\mathbb{Z}[x]$ is not a PID.

(Tut 8, Q3)

Show that $\mathbb{Z}[x]$ is not a PID.

If $\mathbb{Z}[x]$ were a PID, then every ideal would be principal. But consider the ideal $I = \langle 2, x \rangle$. Suppose that I is principal. Then there exists some $\alpha \in \mathbb{Z}[x]$ such that

$$\langle \alpha \rangle = \langle 2, x \rangle.$$

But then this implies that $\alpha \mid 2$ and $\alpha \mid x$. If $\alpha \mid 2$, then $\alpha \in \{\pm 1, \pm 2\}$. If $\alpha \in \{-2, 2\}$, then $\alpha \nmid x$. Hence, $\alpha \in \{-1, 1\}$. If $\alpha = \pm 1$, then $\langle 2, x \rangle$ generates the entire ring. But $3 \notin \langle 2, x \rangle$; hence, $\langle 2, x \rangle$ is not the entire ring. So, the ideal is not principal which means that $\mathbb{Z}[x]$ is not a PID.

Euclidean Domains

We now introduce another algebraic domain that gives us properties of factorisation that are even stronger than principal ideal domains.

Consider a domain D and a function $\nu : D \rightarrow \mathbb{N}$ such that

- for nonzero elements $f, g \in D$, $\nu(f) \leq \nu(fg)$, and
- for any $f \in D$ and $g \in D \setminus \{0\}$, there exist $q, r \in D$ such that $f = gq + r$ with $r = 0$ or $\nu(r) < \nu(g)$.

If such a function exists, then D is called a *Euclidean domain*.

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If such a function exists, then D is called a *Euclidean domain*.

A Euclidean Domain is a PID

It turns out that every Euclidean Domain is a PID (and hence, a UFD). So we have

$$\text{ED} \subseteq \text{PID} \subseteq \text{UFD}$$

Euclidean Domains

Some facts about Euclidean Domains...

Euclidean algorithm

We can apply the Euclidean algorithm to find the greatest common divisor of any two elements.

Quadratic Integer Ring

$\mathbb{Z}[\sqrt{-2}]$ is a ED under $\nu(a + b\sqrt{-2}) = |a^2 + 2b^2|$.

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$\mathbb{Z}[\sqrt{-2}]$ is a ED under $\nu(a + b\sqrt{-2}) = |a^2 + 2b^2|$.

Similar to lectures, proving $\nu(fg) \geq \nu(f)$ is clear as ν is a norm. To prove the second property, $r, p \in \mathbb{Z}[\sqrt{-2}]$, $r = pq + s$, we can choose some $\tilde{q}$ which is as close to $\frac{r}{p}$ by at most $\frac{\sqrt{3}}{2}$, having

$$\nu(s) = |r - pq|^2 = |p|^2 |\tilde{q} - q|^2 = \frac{3}{4} |p|^2 < |p|^2 = \nu(p).$$

Polynomials and EDs

If K is a field, then $K[x]$ is a Euclidean Domain, with $\nu(p) = \deg(p)$.

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Polynomials and EDs

If K is a field, then $K[x]$ is a Euclidean Domain, with $\nu(p) = \deg(p)$.

(2021, Q3 i)

Is $\mathbb{Z}[\sqrt{-5}]$ a Euclidean domain?

We showed that 21 did not factor uniquely. This implies that $\mathbb{Z}[\sqrt{-5}]$ is not a UFD. Since $\mathbb{Z}[\sqrt{-5}]$ is not a UFD, it is not a Euclidean domain.

Gauss' Lemma I

Primitive polynomials

Let R be a UFD and consider some nonzero polynomial $f \in R[x]$. We say that f is *primitive* if the coefficients are all coprime; that is, the greatest common divisor among the coefficients is 1.

Gauss' Lemma I

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Let R be a UFD and consider some nonzero polynomial $f \in R[x]$. We say that f is *primitive* if the coefficients are all coprime; that is, the greatest common divisor among the coefficients is 1.

Gauss' lemma

Let R be a UFD and suppose that f and g are primitive polynomials in $R[x]$. Then the product fg is also primitive.

Let R be a UFD, K be its fraction field. Then every $f \in K[x]$ has a decomposition $f = \lambda g$, $\lambda \in K^\times$, $g \in R[x]$ where g is primitive. This decomposition is unique up to multiplying λ by u and g by u^{-1} where $u \in R^\times$. If $f \in R[x]$, then $\lambda \in R$.

Gauss' Lemma II

Gauss' lemma

Consider some UFD, R . It turns out that the irreducible elements in $R[x]$ are:

- irreducible elements of R , and
- primitive polynomials of positive degree which are irreducible in $K[x]$ where K is the fraction field of R .
- Further, every irreducible element is prime.

UFD Polynomial Ring

UFD Polynomial Ring

Let R be a UFD. Then $R[x]$ is a UFD.

Why? It suffices to show every element f is a product of irreducibles. We can factor f first into product of R and a primitive, then uniquely into irreducibles in $K[x]$ as it is a UFD. By the above, we can choose the irreducibles in $K[x]$ to be primitive irreducibles in $R[x]$ by moving out factors of $K^\times$, then apply Gauss' lemma.

Irreducibility Question

(Pset 9, Q9)

Prove that $\cos(\frac{\pi}{9})$ is a root of $f(x) = 8x^3 - 6x - 1$, and using Gauss' lemma, prove $f(x)$ is irreducible over $\mathbb{Q}[x]$.

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Consider the identity $\cos(3x) = 4\cos(x)^3 - 3\cos(x)$. Now for irreducibility, as f is primitive, proving irreducible over $\mathbb{Q}[x]$ is equivalent to proving irreducibility over $\mathbb{Z}[x]$. Since degree is odd, we must have a linear root, $ax + b$, where a divides 8, b divides -1 , i.e., root must be of form $\pm 1, \pm \frac{1}{2}, \pm \frac{1}{4}, \pm \frac{1}{8}$, but one can check none of them are roots.

6. Fields and Their Extensions

Fields

We now place some more restrictions on the axioms of a ring. Let R be a ring. Recall that

- R under addition is an abelian group.
- R is closed and associative under multiplication.
- The operations are linked by the distributive properties.

For R to be a field, we require the following conditions:

- R contains the multiplicative identity.
- R is commutative under multiplication.
- Every element in R has a multiplicative inverse.

Then R is a field. We usually denote this as F .

Extensions of a field

Let F be a field. What happens if I add in new elements into F ? What sort of elements should I include for the bigger field to remain a field? We will formalise this.

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Extension field

Let $F \subseteq E$ be a field. If E is a field, then we call E an extension field of F . We also say that E/F is a field extension.

Now imagine you have $\alpha_1, \alpha_2, \dots, \alpha_n \in E$. Then $F(\alpha_1, \alpha_2, \dots, \alpha_n)$ is the smallest subfield of E that contains $F, \alpha_1, \alpha_2, \dots, \alpha_n$.

We call an extension *simple* if it is of the form $F(\alpha)$.

Algebraic and transcendental elements

Definition of algebraic and transcendental elements

Let $F \subseteq E$ and consider some element $\alpha \in E$. We call α *algebraic* if α is a root of a (non-zero) polynomial p with coefficients in F . By the First Isomorphism Theorem, $F[\alpha] \cong F[x]/\langle p \rangle$ where $p(x) \in \ker\{\epsilon_\alpha\}$ has minimal degree. We call $p(x)$ the minimal polynomial of α over F . Otherwise, α is said to be *transcendental*.

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Properties

- If $\alpha \in E$ is algebraic over F , then its minimal polynomial is irreducible in $F[x]$.
- Let $E = F(\alpha)$. If α is algebraic over F with minimal polynomial $p \in F[x]$, then $E = F[\alpha] \cong F[x]/\langle p \rangle$. If α is transcendental over F , then $E \cong F(x)$ following the universal property of fraction fields.

Irreducibility of polynomials

Eisenstein's criterion

Let R be a unique factorisation domain with a fraction field K . Let $f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n \in R[x]$. Suppose there exists some prime p such that $p \mid a_0, a_1, \dots, a_{n-1}$ but $p \nmid a_n$ and $p^2 \nmid a_0$. Then f is irreducible over K .

This gives us a sufficient condition to determine if f is irreducible. But failure of the criterion does not guarantee that f is *not* irreducible.

Minimal Polynomial

(2023, Q4 a, b)

Find the minimal polynomial of $\alpha = \sqrt{5 + \sqrt{11}}$ over $\mathbb{Q}$. Prove that the polynomial is minimal.

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We want to construct an irreducible polynomial $p(x)$ with coefficients in $\mathbb{Q}$ so that $p(\alpha) = 0$. We have that

$$\begin{aligned}\alpha = \sqrt{5 + \sqrt{11}} &\implies \alpha^2 = 5 + \sqrt{11} \\ &\implies \alpha^2 - 5 = \sqrt{11} \\ &\implies \alpha^4 - 10\alpha^2 + 25 = 11 \\ &\implies \alpha^4 - 10\alpha^2 + 14 = 0.\end{aligned}$$

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To prove that it is minimal, we use Eisenstein's criterion. We have that $2 \mid 10, 2 \mid 14$, yet $2 \nmid 1$ and $2^2 \nmid 14$. Hence, the polynomial is irreducible and therefore minimal.

Irreducibility

(Tut 10, Q7)

Let $F = \mathbb{F}_p(x)$, and consider the polynomial $f = y^p - x \in F[y]$. Prove that f is irreducible.

Irreducibility

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Note since f is primitive over the field $\mathbb{F}_p[x]$ and F is the fraction field of $\mathbb{F}_p[x]$, by Gauss' lemma, it suffices to prove f is irreducible over $\mathbb{F}_p[x][y]$. Then we clearly have $x \mid x, x \nmid 1$ and $x^2 \nmid x$, whence by Eisenstein, we have f is irreducible.

Finite Extensions and their Degree

Degree of an Extensions

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Finite Extensions and their Degree

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Let $F \subseteq E$ be a field. Then E can be viewed as a “vector space” over F with the degree of the extension being the *dimension* of E over F .

Now suppose that α is algebraic over some field F . Then the degree of the extension field $F(\alpha)$ over F is precisely the degree of the minimal polynomial; that is,

$$[F(\alpha) : F] = \deg p,$$

where p is the minimal polynomial of α over F .

In fact, we can construct a tower of extensions. Suppose that we have fields F, E, H satisfying $E \subseteq F \subseteq H$. Then

$$[H : F] = [H : E] \cdot [E : F].$$

(Pset 9, Q9)

As from our previous example, since we showed the minimal polynomial for $\cos(\frac{\pi}{9})$ is $8x^3 - 6x - 1$, we can conclude that $[\mathbb{Q}(\cos(\frac{\pi}{9}) : \mathbb{Q}] = 3$.

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(2020, Q5 ii)

Let $p \geq 3$ and define $\theta = 1 + \sqrt[p]{p} + \sqrt[p]{p^3} + \cdots + \sqrt[p]{p^{2p+1}}$. Find $[\mathbb{Q}(\theta) : \mathbb{Q}]$.

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Observe that

$$\mathbb{Q} \subseteq \mathbb{Q}(\theta) \subseteq \mathbb{Q}(\sqrt[p]{p}).$$

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Observe that

$$\mathbb{Q} \subseteq \mathbb{Q}(\theta) \subseteq \mathbb{Q}(\sqrt[p]{p}).$$

We, therefore, have that

$$[\mathbb{Q}(\theta) : \mathbb{Q}] \mid [\mathbb{Q}(\sqrt[p]{p}) : \mathbb{Q}].$$

By Eisenstein's criterion on p , we claim that $x^p - p$ is the minimal polynomial of $\sqrt[p]{p}$ over $\mathbb{Q}$. In other words, we have that

$$[\mathbb{Q}(\theta) : \mathbb{Q}] \mid p \implies [\mathbb{Q}(\theta) : \mathbb{Q}] = \{1, p\} \implies [\mathbb{Q}(\theta) : \mathbb{Q}] = p.$$

Splitting fields

Let $f(x) = x^2 - 8$. If we consider this polynomial over $\mathbb{Q}$, then f is irreducible because the roots $\sqrt{8}$ and $-\sqrt{8}$ do not belong in $\mathbb{Q}$.

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$$f(x) = (x - 2\sqrt{2})(x + 2\sqrt{2})$$

and each of these coefficients belong in $\mathbb{Q}(\sqrt{2})$. We call $\mathbb{Q}(\sqrt{2})$ the *splitting field* of f because it splits f into linear factors.

Splitting fields

More generally,

Splitting field

Let F be a field. A *splitting field* of some polynomial $p(x)$ over F is an extension field K such that p splits into linear factors.

We generally restrict ourselves to the *smallest* field for which this happens and call this *the splitting field* of p .

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Kronecker

Let F be a field and consider $f \in F[x]$. Then there exist a finite extension E/F such that f is a product of linear factors in $E[x]$; that is,

$$f(x) = a(x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_n),$$

where $\alpha_1, \alpha_2, \dots, \alpha_n \in E$.

Extension Degree

(2021, Question 5 ii)

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Recall that the splitting field separates the polynomial into linear factors. So the splitting field must contain all of its roots. The roots of $f(x)$ are $\{\sqrt[3]{11}, \sqrt[3]{11}\xi, \sqrt[3]{11}\xi^2\}$ where ξ denotes the primitive root of degree 3. The splitting field, therefore, becomes $\mathbb{Q}(\sqrt[3]{11}, \xi)$.

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To determine its degree, we can compute the tower of extensions

$$[\mathbb{Q}(\sqrt[3]{11}, \xi) : \mathbb{Q}] = [\mathbb{Q}(\sqrt[3]{11})(\xi) : \mathbb{Q}(\sqrt[3]{11})] \cdot [\mathbb{Q}(\sqrt[3]{11}) : \mathbb{Q}].$$

The minimal polynomial of $\mathbb{Q}(\sqrt[3]{11}, \xi)$ over $\mathbb{Q}(\sqrt[3]{11})$ has degree 2 while the minimal polynomial of $\mathbb{Q}(\sqrt[3]{11})$ over $\mathbb{Q}$ has degree 3. Thus, $\mathbb{Q}(\sqrt[3]{11}, \xi)$ is an extension of degree $2 \times 3 = 6$.

Finite fields

Let F be a field. We call F a *finite field* if it has finitely many elements.

Finite Fields

We note any field must have characteristic 0 or a prime. For finite fields, the former clearly can't hold true. If F is a finite field with characteristic p , then we have F contains $\mathbb{F}_p$, and $n = [F : \mathbb{F}_p]$, $|F| = p^n$ since $F \cong (\mathbb{F}_p)^n$ as a vector space.

Finite fields

Structure of Finite Fields

Let us have $q = p^n$ where p is prime. Consider $f = x^q - x \in \mathbb{F}_p[x]$. There exists a field extension $E/\mathbb{F}_p$ such that f splits into product of linear factors in $E[x]$.

$$F = \{\alpha \in E \mid f(\alpha) = 0\}.$$

We have F is a field with exactly $q = p^n$ elements. We also have that if two fields are both of order q , then they are isomorphic.

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(Pset 9, Q18)

Let $f \in \mathbb{F}_p[x]$ be a monic irreducible polynomial of degree n . Prove that $\mathbb{F}_{p^n}$ is the splitting field of f .

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(Pset 9, Q18)

Let $f \in \mathbb{F}_p[x]$ be a monic irreducible polynomial of degree n . Prove that $\mathbb{F}_{p^n}$ is the splitting field of f .

Since f is irreducible and $\mathbb{F}_p[x]$ is a PID (ED), we have $\langle f \rangle$ is maximal. Hence $K = \mathbb{F}_p[x]/\langle f \rangle \cong \mathbb{F}_p(\alpha)$ where $\alpha = x + \langle f \rangle$ is a field. Due to the max degree being less than n with p choices for each coefficient, we have $|K| = p^n$, i.e., $K \cong \mathbb{F}_{p^n}$.

Note by a lemma in the lectures, we have f must divide $x^q - x$, implying all roots of f are also roots of $x^q - x$. Since K has q elements, $|K^\times| = q - 1$ which implies all elements of K are roots to $x^q - x$. Since $x^q - x$ has at most q roots, we have K are all the roots. Hence it has all roots of f , implying $K \cong \mathbb{F}_{p^n}$ is the splitting field.

Finite fields

(2020, Q4 iii)

Prove that the rings $(\mathbb{Z}/3\mathbb{Z})[x]/(x^2 + x + 2)$ and $(\mathbb{Z}/3\mathbb{Z})[x]/(x^2 + 2x + 2)$ are isomorphic.

Finite fields

(2020, Q4 iii)

Prove that the rings $(\mathbb{Z}/3\mathbb{Z})[x]/(x^2 + x + 2)$ and $(\mathbb{Z}/3\mathbb{Z})[x]/(x^2 + 2x + 2)$ are isomorphic.

It is sufficient to prove that these are finite fields of the same order. This amounts to showing that $x^2 + x + 2$ and $x^2 + 2x + 2$ are irreducible over $\mathbb{Z}_3[x]$, since the two polynomials have the same degree.

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We first prove that $x^2 + x + 2$ is irreducible over $\mathbb{Z}_3[x]$, meaning that $x^2 + x + 2$ has no roots. We have that

$$0^2 + 0 + 2 = 2 \neq 0,$$

$$1^2 + 1 + 2 = 1 \neq 0,$$

$$2^2 + 2 + 2 = 2 \neq 0.$$

Finite Fields

(2020, Q4 iii)

Prove that the rings $(\mathbb{Z}/3\mathbb{Z})[x]/(x^2 + x + 2)$ and $(\mathbb{Z}/3\mathbb{Z}[x])/(x^2 + 2x + 2)$ are isomorphic.

Similarly $x^2 + 2x + 2$ is irreducible over $\mathbb{Z}_3[x]$, since

$$0^2 + 2 * 0 + 2 = 2 \neq 0,$$

$$1^2 + 2 * 1 + 2 = 2 \neq 0,$$

$$2^2 + 2 + 2 = 1 \neq 0.$$

So the fields are both finite fields of order 9, and hence they must be isomorphic.

Finite Fields

(2015, Question 5)

Consider the following rings, each of which has 9 elements:

- $\mathbb{Z}/9\mathbb{Z}$,
- $\mathbb{F}_9$,
- $\mathbb{F}_3 \times \mathbb{F}_3$,
- $\mathbb{F}_3[x]/(x^2)$,
- $\mathbb{F}_3[x]/(x^2 + 1)$,
- $\mathbb{F}_3[x]/(x^2 - 1)$.

Sort these rings into equivalence classes under isomorphism.

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- $\mathbb{F}_3[x]/(x^2 - 1)$.

Sort these rings into equivalence classes under isomorphism.

First of all, $x^2 + 1$ is irreducible over $\mathbb{F}_3$ since

$$0^2 + 1 = 1 \neq 0, \quad 1^2 + 1 = 2 \neq 0, \quad \text{and} \quad 2^2 + 1 = 2 \neq 0.$$

So $\mathbb{F}_3[x]/(x^2 + 1) \sim \mathbb{F}_9$.

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- $\mathbb{F}_3[x]/(x^2 + 1)$,
- $\mathbb{F}_3[x]/(x^2 - 1)$.

To understand the structure of $\mathbb{F}_3[x]/(x^2 - 1)$, we use the first isomorphism theorem for rings. That is, since the kernel of the mapping $p \rightarrow (p(2), p(1))$ has kernel $(x^2 - 1)$,

$$\mathbb{F}_3[x]/(x^2 - 1) \cong \mathbb{F}_3 \times \mathbb{F}_3.$$

Finite Fields

We note that $\mathbb{F}_3 \times \mathbb{F}_3$ is not isomorphic to $\mathbb{F}_9$ since the latter is a field and the former is not, so this gives us two equivalence classes already.

Finite Fields

(2015, Question 5)

Consider the following rings, each of which has 9 elements:

- $\mathbb{Z}/9\mathbb{Z}$,
- $\mathbb{F}_9$,
- $\mathbb{F}_3 \times \mathbb{F}_3$,
- $\mathbb{F}_3[x]/(x^2)$,
- $\mathbb{F}_3[x]/(x^2 + 1)$,
- $\mathbb{F}_3[x]/(x^2 - 1)$.

Consider $\mathbb{Z}/9\mathbb{Z}$. It has an element of (additive) order 9 and is not a field, so it must be in a distinct equivalence class from the other rings.

Finite Fields

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Consider $\mathbb{Z}/9\mathbb{Z}$. It has an element of (additive) order 9 and is not a field, so it must be in a distinct equivalence class from the other rings.

Finite Fields

Furthermore, $\mathbb{F}_3[x]/(x^2)$ possesses a non-zero nilpotent element x with order 2, which, of the other rings, is only true of $\mathbb{Z}/9\mathbb{Z}$. However, it does not have an element with (additive) order 9, so it must be distinct from $\mathbb{Z}/9\mathbb{Z}$. In conclusion, the equivalence classes are

$$\begin{aligned} &\{\mathbb{Z}/9\mathbb{Z}\}, \\ &\{\mathbb{F}_3[x]/(x^2)\}, \\ &\{\mathbb{F}_3[x]/(x^2 + 1), \mathbb{F}_9\}, \text{ and} \\ &\{\mathbb{F}_3[x]/(x^2 - 1), \mathbb{F}_3 \times \mathbb{F}_3\}. \end{aligned}$$

Finite Fields

(2022, Question 5 i)

Let α and β be complex numbers. Prove that if $\alpha + \beta$ and $\alpha\beta$ are algebraic over $\mathbb{Q}$, then α and β are also algebraic over $\mathbb{Q}$.

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$$[\mathbb{Q}(\alpha + \beta, \alpha\beta) : \mathbb{Q}]$$

is finite since $\alpha + \beta$ and $\alpha\beta$ are algebraic over $\mathbb{Q}$.

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is finite since $\alpha + \beta$ and $\alpha\beta$ are algebraic over $\mathbb{Q}$. Now

$$x^2 - (\alpha + \beta)x + \alpha\beta$$

is a polynomial with roots α and β . Since $(\alpha + \beta) - \alpha = \beta$, we have

$\mathbb{Q}(\alpha + \beta, \alpha\beta, \alpha) = \mathbb{Q}(\alpha + \beta, \alpha\beta, \alpha, \beta)$. Hence

$[\mathbb{Q}(\alpha + \beta, \alpha\beta, \alpha, \beta) : \mathbb{Q}(\alpha + \beta, \alpha\beta)] \leq 2$, and by the tower of extensions, $[\mathbb{Q}(\alpha + \beta, \alpha\beta, \alpha, \beta) : \mathbb{Q}]$ is finite. So α and β are algebraic over $\mathbb{Q}$.

UNSW Mathematics Society $\times$ Competitive
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MATH3711/5706 Seminar



Presented by Jeff, Brendan, Justin and Daniel

Attendance

Please scan the QR code; please let us know about any dietary requirements (for the pizza!).



1. Introduction to Group Theory

Basic Notions

Group

A group G is a set equipped with a associative and invertible binary operation that has an identity. If the operation is commutative then G is **abelian**. We call the cardinality of the set the **order** of the group.

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Generation and Order

Order

For each $g \in G$, the *order* of g is the least positive integer n such that

$$g^n = 1.$$

In this case, we write $\text{ord}(g) = n$. If no such integer exists, the element is said to have *infinite order*.

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Generation

If $S \subseteq G$, then $\langle S \rangle$ is the smallest subgroup of G containing S , given by

$$\langle S \rangle = \bigcap_{\substack{H \leq G \\ S \subseteq H}} H = \{s_1 s_2 \cdots s_n : s_i \in S \cup S^{-1}, n \geq 0\}.$$

A group is said to be *finitely generated* if it is generated by a finite set.

Group Question

Group Proof Example

Let G be a group and $u \in G$. If we define a new multiplication on G by the rule: $a \circ b = au^{-1}b$ for all $a, b \in G$. Show that G together with this new multiplication forms a group.

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For associativity we see that with $a, b, c \in G$

$$(a \circ b) \circ c = (au^{-1}b) \circ c = au^{-1}bu^{-1}c = au^{-1}(bu^{-1}c) = a \circ (b \circ c).$$

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We can use $u \in G$ as an identity element since

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Any element $g \in G$ has the inverse $ug^{-1}u \in G$ where

$$g \circ (ug^{-1}u) = gu^{-1}ug^{-1}u = u = ug^{-1}uu^{-1}g = (ug^{-1}u) \circ g.$$

A Proof Involving Subgroups

Example

Let G be a group with an element g such that $H = G \setminus \{g\}$ is a subgroup of G . Show that $|G| = 2$.

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which implies $h = 1$.

Thus $H = \{1\}$, so $|H| = 1$ and therefore $|G| = 2$. □

Justin's Favorite Groups

Useful groups that we have covered in this course to go to for examples:

- $\mathbb{Q}, \mathbb{Z}, \mathbb{R}, \mathbb{C}$ under addition
- $\mathbb{Q}^\times = \mathbb{Q} \setminus \{0\}$ under multiplication, and likewise for $\mathbb{R}^\times = \mathbb{R} \setminus \{0\}$, $\mathbb{C}^\times = \mathbb{C} \setminus \{0\}$ and $\mathbb{Z}^\times = \{1, -1\}$
- $\mathbb{Z}_n (= \mathbb{Z}/n\mathbb{Z})$, the group of $\{0, 1, \dots, n-1\}$ under addition
- For a ring R , $GL_n(R)$ (under matrix addition)
- The dihedral group D_n , the symmetric group S_n , and the alternating group A_n (see next slides)
- The trivial group consisting of one element e

And quotients and direct products of these are the common groups you should know.

Detour on S_n and D_n

Know how to work with S_n and D_n , and use examples from them.

Symmetric Group

Let $S = \{1, 2, \dots, n\}$. Define $S_n := \text{Perm}(S)$ to be the set of permutations $\sigma : S \rightarrow S$ (i.e. bijections). This set forms a group under composition $\circ$, called the symmetric group on n letters.

Dihedral Group

Define

$$D_n = \langle \sigma, \tau \mid \tau^2 = e, \sigma^n = e, \tau\sigma = \sigma^{-1}\tau \rangle$$

This set can be expressed as

$$D_n = \{e, \sigma, \sigma^2, \dots, \sigma^{n-1}, \tau, \sigma\tau, \sigma^2\tau, \dots, \sigma^{n-1}\tau\},$$

the group of symmetries of the n -gon, with the given relators.

Sign in S_n

Recall

- A transposition is a cycle of order 2.
- Any permutation $\sigma \in S_n$ can be written as a product of transpositions.
- Define $s(\sigma) = 1$ if the number of transpositions is even, and $s(\sigma) = -1$ otherwise.
- If $\sigma, \tau \in S_n$, then $s(\sigma\tau) = s(\sigma)s(\tau)$.
- If $\sigma = \tau_1\tau_2 \cdots \tau_k$, then $s(\sigma) = (-1)^k$.
- $A_n = \{\sigma \in S_n : s(\sigma) = 1\}$ is the alternating group.

Cosets

If $H \leq G$, the left coset of $g \in G$ is defined as

$$gH = \{gh : h \in H\}.$$

The set of left cosets is denoted G/H , and if G is finite, Lagrange's Theorem gives

$$[G : H] = |G/H| = \frac{|G|}{|H|}.$$

We can also define $JK := \{jk : j \in J, k \in K\}$.

Example: $\mathbb{Z}/m\mathbb{Z} = \{\overline{0}, \overline{1}, \overline{m-1}\}$.

Normal Subgroups

Equivalent Conditions

We say $N \trianglelefteq G$ is normal if any of the following hold:

- $gN = Ng$ for all $g \in G$
- $gNg^{-1} = N$ for all $g \in G$
- $gNg^{-1} \subseteq N$ for all $g \in G$

Quotient Group Properties

If N is normal then G/N forms a group:

- $(gN)(hN) = (gh)N$
- $1_{G/N} = N$
- $(gN)^{-1} = g^{-1}N$

2. Group Homomorphisms

Homomorphism Definition

Homomorphisms

For groups G, H , a function $\phi : G \rightarrow H$ is a group homomorphism if for all $g, g' \in G$, we have $\phi(gg') = \phi(g)\phi(g')$.

- Bijective $\iff$ Isomorphism
- Injective $\iff$ Monomorphism
- Surjective $\iff$ Epimorphism

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- Surjective $\iff$ Epimorphism

Kernel & Image

There are two important subgroups associated with a homomorphism ϕ : $\ker \phi = \phi^{-1}(1)$ and $\text{im } \phi = \{\phi(g) : g \in G\}$

Homomorphism Example

Original Example

Let $\varphi : G \rightarrow \mathbb{C}^*$ be a nontrivial group homomorphism with G finite. Show that

$$\sum_{g \in G} \varphi(g) = 0.$$

(Hint: as a coset, $hG = G$ for all $h \in G$.)

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Proof. Since φ is nontrivial, there exists $h \in G$ such that $\varphi(h) \neq 1$. Then

$$\sum_{g \in G} \varphi(g) = \sum_{g \in hG} \varphi(g) = \sum_{g \in G} \varphi(hg) = \sum_{g \in G} \varphi(h)\varphi(g) = \varphi(h) \sum_{g \in G} \varphi(g).$$

Hence,

$$(1 - \varphi(h)) \sum_{g \in G} \varphi(g) = 0.$$

Since $\varphi(h) \neq 1$, it follows that $\sum_{g \in G} \varphi(g) = 0$. □

More Definitions

Morphisms

An *automorphism* of G is an isomorphism $\phi : G \rightarrow G$.

It is called *inner* if it is of the form $\phi_g(x) = gxg^{-1}$ for some fixed $g \in G$.

Normalizers

For a subgroup $H \leq G$, define

$$N_G(H) = \{g \in G : gH = Hg\}.$$

Centre

The centre of a group, $Z(G)$, is the set of elements that commute with everything; that is,

$$Z(G) = \{g \in G : gh = hg \ \forall h \in G\}.$$

Quotient and Inclusion Morphisms

Quotient and Inclusion Morphisms

If $N \trianglelefteq G$, then the quotient morphism

$$\pi_N : G \rightarrow G/N, \quad \pi_N(g) = gN$$

is a group homomorphism.

If $H \leq G$, then the inclusion map

$$\eta : H \hookrightarrow G, \quad \eta(h) = h$$

is a monomorphism. Though simple, inclusions are useful when constructing more complicated morphisms.

A Proof Involving Kernels

Original Example

Let G be an abelian group. Show that every subgroup of G is the kernel of a homomorphism from G .

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Let G be an abelian group. Show that every subgroup of G is the kernel of a homomorphism from G .

Proof. Let $H \leq G$. Since G is abelian, every subgroup is normal, so $H \trianglelefteq G$ and G/H is a group.

Consider the quotient map

$$\pi_H : G \rightarrow G/H, \quad \pi_H(g) = gH.$$

Then π_H is a homomorphism. We compute its kernel:

$$\ker(\pi_H) = \{g \in G : gH = H\}.$$

If $g \in \ker(\pi_H)$, then $gH = H$, so in particular $g \in H$ (since $1 \in H$ implies $g = gh$ for some $h \in H$, hence $g \in H$).

A Proof Involving Kernels

Conversely, if $h \in H$, then $hH = H$, so $h \in \ker(\pi_H)$, hence $H \subseteq \ker(\pi_H)$.

Therefore,

$$\ker(\pi_H) = H.$$

Thus $\ker(\pi_H) \subseteq H$.

Isomorphism Theorems

First Isomorphism Theorem

If $\phi : G \rightarrow H$ then $G/\ker(\phi) \cong \text{im}(\phi)$.

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Second Isomorphism Theorem

If $H \leq G$ and $N \trianglelefteq G$, then $HN \leq G$ and $H \cap N \trianglelefteq H$, and $\frac{HN}{N} \cong \frac{H}{H \cap N}$.

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Third Isomorphism Theorem

If $N \leq H \trianglelefteq G$ and $N \trianglelefteq G$, then $\frac{G/N}{H/N} \cong G/H$.

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If $N \leq H \trianglelefteq G$ and $N \trianglelefteq G$, then $\frac{G/N}{H/N} \cong G/H$.

Fourth Isomorphism Theorem

Let $H \trianglelefteq G$. Then the set of subgroups containing H is in bijection with the subgroups of G/H . moreover this bijection preserves normal subgroups.

Using the Isomorphism Theorems

Original Example

Prove that there is no nontrivial homomorphism

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when m and n are coprime.

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when m and n are coprime.

Proof. Let $\varphi : \mathbb{Z}/m\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ be a homomorphism, and let $r = |\operatorname{im} \varphi|$. On one hand, by the First Isomorphism Theorem,

$$\frac{|\mathbb{Z}/m\mathbb{Z}|}{|\ker \varphi|} = |\operatorname{im} \varphi| = r,$$

so $r \mid m$.

On the other hand, since $\operatorname{im} \varphi \leq \mathbb{Z}/n\mathbb{Z}$, Lagrange's theorem gives $r \mid n$. Since m and n are coprime, it follows that $r = 1$, hence $\operatorname{im} \varphi$ is trivial and φ is the trivial homomorphism. $\square$

Using the Isomorphism Theorems Again

Adapted Example

Given that

$$N := \{\lambda I : \lambda \in \mathbb{C}^*\} \trianglelefteq GL_n(\mathbb{C}),$$

find a quotient group G of $SL_n(\mathbb{C})$ such that

$$G \cong GL_n(\mathbb{C})/N.$$

Using the Isomorphism Theorems Again

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By the Third Isomorphism Theorem,

$$\frac{SL_n(\mathbb{C})N}{N} \cong \frac{SL_n(\mathbb{C})}{SL_n(\mathbb{C}) \cap N}.$$

It remains to identify $SL_n(\mathbb{C})N$ and $SL_n(\mathbb{C}) \cap N$.

First, we show $SL_n(\mathbb{C})N = GL_n(\mathbb{C})$.

Clearly $SL_n(\mathbb{C})N \subseteq GL_n(\mathbb{C})$. For the reverse inclusion, let $M \in GL_n(\mathbb{C})$ with $\det M = d \neq 0$. Set $\lambda \in \mathbb{C}^*$ such that $\lambda^n = d$.

Adapted Example

Then

$$M = (\lambda^{-1}M)(\lambda I),$$

where $\lambda I \in N$ and

$$\det(\lambda^{-1}M) = \lambda^{-n} \det M = 1,$$

so $\lambda^{-1}M \in SL_n(\mathbb{C})$.

Thus $GL_n(\mathbb{C}) \subseteq SL_n(\mathbb{C})N$, and hence

$$SL_n(\mathbb{C})N = GL_n(\mathbb{C}).$$

Next,

$$SL_n(\mathbb{C}) \cap N = \{\omega I : \det(\omega I) = 1\} = \{\omega I : \omega^n = 1\}.$$

Therefore,

$$G \cong \frac{SL_n(\mathbb{C})}{\{\omega I : \omega^n = 1\}}.$$

3. Direct Products and Sums

External Direct Products

Definition (External direct product)

Suppose we have groups $G_1, G_2, \dots, G_n$. The Cartesian product of these groups, with the operation of coordinatewise multiplication, is also a group, called the **external direct product** of the groups.

Quick example

Consider the group $G = (\mathbb{R}^+, *)$. $G \times G$, taken as an external direct product, is simply $\mathbb{R}^\times \times \mathbb{R}^\times$. That is, $(2, 3) * (4, 5) = (8, 15)$.

External Direct Products

Definition (Canonical projections and injections)

Two important morphisms associated with the external direct product are the **canonical projection** and the **canonical injection**.

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Two important morphisms associated with the external direct product are the **canonical projection** and the **canonical injection**.

1. **Canonical projection** $\pi_k : G_1 \times G_2 \times \dots \times G_k \times \dots \times G_n \rightarrow G_k$
such that

$$\pi_k : (g_1, g_2, \dots, g_k, \dots, g_n) \mapsto g_k.$$

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such that

$$\pi_k : (g_1, g_2, \dots, g_k, \dots, g_n) \mapsto g_k.$$

It 'projects' the direct product onto a single group. It is an epimorphism.

External Direct Products

Definition (Canonical projections and injections)

Two important morphisms associated with the external direct product are the **canonical projection** and the **canonical injection**.

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It is so named because it is injective (so a monomorphism). We do not 'lose information' through this map, but the non- k th coordinates are all set to the identity.

Quotients are Just Fractions?

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and compare it with $\mathbb{Z}$.

The group $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$ contains an element of order m , for instance

$$(1 + m\mathbb{Z}, 0),$$

whereas $\mathbb{Z}$ has no nontrivial finite-order elements.

Since isomorphisms preserve orders of elements, it follows that

$$\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z} \not\cong \mathbb{Z}.$$



Be Careful

Unlike the previous slide, the following is true.

Result

Interpreting $1 \times H$ as

$$\{(1_G, h) : h \in H\},$$

it is true that

$$(G \times H)/(1 \times H) \cong G.$$

However, it is not in general true that

$$G/H \times H \cong G.$$

Internal Direct Products

Definition (Internal direct product)

Consider groups $G_1, G_2, \dots, G_n \leq G$ where

1. the elements of the different groups commute, but the individual groups are not necessarily abelian, and
2. for all $i = 1, 2, \dots, n$, $G_i \cap (G_1 G_2 \dots G_{i-1} G_{i+1} \dots G_n) = \{1_G\}$. That is, it is impossible to get any non-identity element in one subgroup from multiplying elements in the other subgroups together.

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Then the group $G_1 G_2 \dots G_n$, termed the **internal direct product**, is isomorphic to the **external direct product** $G_1 \times G_2 \times \dots \times G_n$.

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Then the group $G_1 G_2 \dots G_n$, termed the **internal direct product**, is isomorphic to the **external direct product** $G_1 \times G_2 \times \dots \times G_n$.

Intuitively, this is because the homomorphism

$\phi : G_1 \times G_2 \times \dots \times G_n \rightarrow G_1 G_2 \dots G_n$ where

$$\phi(g_1, g_2, \dots, g_n) = g_1 g_2 \dots g_n$$

is a bijection.

Direct Sum of Subgroups

Let $G_1, G_2, \dots, G_n$ be subgroups of the abelian group G that generate G . Then G is the direct sum of $G_1, G_2, \dots, G_n$, written as

$$G = G_1 \oplus G_2 \oplus \cdots \oplus G_n$$

if and only if the only solution to

$$a_1 + a_2 + \cdots + a_n = 0, \quad \text{where } a_i \in G_i, \ i = 1, 2, \dots, n,$$

is

$$a_1 = a_2 = \cdots = a_n = 0.$$

For finite groups, the direct sum coincides with the direct product.

Finite Abelian Groups

Cyclic Groups

The only cyclic groups up to isomorphism are $\mathbb{Z}/n\mathbb{Z}$ for $n \geq 1$.

Finite Abelian Groups

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The only cyclic groups up to isomorphism are $\mathbb{Z}/n\mathbb{Z}$ for $n \geq 1$.

Structure Theorem for Finite Abelian Groups

A finite abelian group is isomorphic to a direct sum of cyclic groups. More precisely, if A is a finite abelian group, then there exist positive integers $m_1, \dots, m_k$, each a prime power, such that

$$A \cong \mathbb{Z}/m_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/m_k\mathbb{Z}.$$

Further, the sequence $m_1, \dots, m_k$ is unique up to reordering.

Finite Abelian Groups: Putting it all Together

For instance, how might we use this to find all the finite abelian groups of order 360 up to isomorphism?

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First, we will decompose 360 into its prime factors that is

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Finite Abelian Groups: Putting it all Together

For instance, how might we use this to find all the finite abelian groups of order 360 up to isomorphism?

First, we will decompose 360 into its prime factors that is

$$360 = 2^3 \times 3^2 \times 5.$$

The possibilities are:

$$\mathbb{Z}/360\mathbb{Z}$$

$$\mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/60\mathbb{Z}$$

$$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/180\mathbb{Z}$$

$$\mathbb{Z}/3 \times \mathbb{Z}/120\mathbb{Z}$$

$$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/90\mathbb{Z}$$

$$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/30\mathbb{Z}.$$

4. Group Actions & Sylow Theorems

G -sets and group actions

We can start to think about a group 'acting' on elements in a given set.

Definition (G -set)

Suppose we have a set S , and we define a map $\alpha : G \times S \rightarrow S$ (where we can define $\alpha(g, s) := g.s$). Then we can say that G acts on the set S , denoted $G \curvearrowright S$, so long as the following conditions hold:

1. $(gh).s = g.(h.s)$ for all $g, h \in G$ and $s \in S$.
2. $1_G.s = s$ for all $s \in S$

The set together with the operation is called a G -set.

We can think of this as a group homomorphism from G to $\text{Perm}(S)$, since for each element of G we define an invertible action from S to itself.

G -sets and group actions

Examples (G -sets)

1. The set G is a G -set under the group action $g.g' = gg'$.
2. The set G is also a G -set under the conjugation action $g.n = gng^{-1}$.
3. The set G/H is a G -set under the group action $g'.gH = (g'g)H$
4. The trivial action $G \curvearrowright S$ where $g \cdot s = s$ for all $g \in G$ and $s \in S$.

Definition (G -stable subsets)

A subset $T \subseteq S$ is G -stable if $G.T \subseteq T$.

Orbits

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A subset $T \subseteq S$ is G -stable if $G.T \subseteq T$.

Definition (Orbit)

The orbit of $s \in S$ is the subset $G.s$. It is the smallest G -stable subset of S that contains s . It can be thought of as all the elements of S that one can 'get to' from acting on s .

Orbits

Theorem (Orbit structures)

If we define the relation $\sim$ as $s \sim s'$ if and only if $s \in G.s'$, then $\sim$ is an equivalence relation. Therefore, the G -orbits of a set form a partition of the set. This also implies that G -orbits are disjoint. We denote S/G as the set of orbits from the action $G \curvearrowright S$.

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A group G acts transitively on a set S if S consists of only one orbit.

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Definition (Transitive action)

A group G acts transitively on a set S if S consists of only one orbit.

Definition (Faithful action)

A group G acts faithfully if $g \cdot s = s$ for all $s \in S$ implies that g is the identity.

G -set Morphisms

Isomorphism of G -sets

For two G -sets S_1 and S_2 , a G -set morphism is a function $\psi : S_1 \rightarrow S_2$ such that

$$\psi(g.s) = g.\psi(s) \text{ for all } g \in G, s \in S.$$

If ψ is bijective, then the G -sets S_1 and S_2 are isomorphic.

Stabilisers

Definition (Stabilisers)

For S a G -set, the stabiliser of the element $s \in S$ is the subgroup

$$\text{stab}_G(s) = \{g \in G : g.s = s\}.$$

That is, it is the subgroup of all elements g that fix $s \in S$.

Common Orbits and Stabilizers

Examples

- (a) $G \curvearrowright G/H$ where $H \leq G$, defined by $g' \cdot (gH) = g'gH$. This action is transitive, and $\text{Stab}(gH) = g \text{Stab}(H) g^{-1} = gHg^{-1}$.

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- (b) $G \curvearrowright G$ where $g \cdot s = gsg^{-1}$. Here the orbits are conjugacy classes. The stabiliser is

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- (c) $G \curvearrowright X$ where $X = \{H \leq G\}$ and $g \cdot H = gHg^{-1}$. The stabiliser is the normaliser of H , denoted $N_G(H)$ where

$$N_G(H) = \{g \in G : gHg^{-1} = H\}.$$

Orbit-stabiliser Theorem

Theorem (Structure of G-Orbits)

Let a group G act transitively on a set S . Let $s \in S$ and $H = \text{stab}_G(s) \leq G$. Then we have the following well defined isomorphism of G -sets.

$$\psi : G/H \rightarrow S; gH \mapsto g.s$$

Corollary (Lagrange's Theorem)

For a G -set S and an element $s \in S$, $|G.s| = |G|/|\text{stab}_G(s)|$. That is,

$$\text{Size of stabiliser} \times \text{Size of orbit} = \text{Order of group}.$$

Orbit-stabiliser Theorem

Original example

Let H and K be two subgroups of the finite group G . $H \times K$ acts on HK according to the map

$$(h_1, k_1).hk = h_1hkk_1^{-1} \text{ for all } h, h_1 \in H, k, k_1 \in K.$$

Use the orbit-stabiliser theorem to show that

$$|HK| = \frac{|H||K|}{|H \cap K|}.$$

We want to be using the theorem $|G.s| = |G|/|\text{stab}_G(s)|$. To do this, we need to identify our group $H \times K$ and our set HK .

We also need to identify a suitable element to calculate the orbit of.

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which is simply $\{(h_1, h_1) \in H \times K : h_1 \in H \cap K\}$ and has size $|H \cap K|$.

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Stabilisers of “conjugate” points

Proposition

Let S be a G -set, $s \in S$, and $g \in G$. Then,

$$\text{Stab}_G(g.s) = g(\text{Stab}_G(s))g^{-1}.$$

Nice Result

Let $H_1, H_2 \leq G$ then $H_1, H_2 \leq G$ are conjugate (i.e., $H_2 = gH_1g^{-1}$). if and only if G/H_1 and G/H_2 are isomorphic as G -sets.

Subgroup Theorems

Suppose from here that p is a prime.

Cauchy's theorem

Let G be a finite group, and suppose that $p \mid |G|$. Then G has an element of order p .

Cayley's theorem

If $G \curvearrowright S$, then there is an induced homomorphism $G \rightarrow \text{Perm}(S)$. Specifically, when $S = G$ with the left multiplication action, this is injective (i.e., the action is faithful), and hence any finite group is isomorphic to a subgroup of S_n for some $n \in \mathbb{N}$.

Counting Orbits

Definition (Fixed point set)

Consider $H \subset G$ and S a G -set. The fixed point set of H is

$$S^H = \{s \in S : h.s = s \text{ for any } h \in H\}.$$

Theorem (Burnside's Lemma)

Suppose that the group G and the G -set S are finite. Then the number of G -orbits in the S is given by

$$|S/G| = \frac{1}{|G|} \sum_{g \in G} |S^g|.$$

Counting Orbits Example

(2020, Question 2 iii)

Let G be a finite group acting transitively upon a finite set with more than one element. Show that there is an element $g \in G$ without fixed points.

Counting Orbits Example

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Let G be a finite group acting transitively upon a finite set with more than one element. Show that there is an element $g \in G$ without fixed points.

If G acts transitively on the set S , then $|S/G| = 1$. Hence,

$$1 = \frac{1}{|G|} \sum_{g \in G} |S^g|.$$

Now our aim is to show that for at least one g , $|S^g| = 0$. Suppose that, instead, $|S^g| \geq 1$ for all g . But we know that for the identity element 1_G , $|S^{1_G}| = \{s \in S : 1_G.s = s\} = |S| > 1$, since S has more than one element. So it follows that

$$\sum_{g \in G} |S^g| > \underbrace{(1 + 1 + \dots + 1)}_{|G| \text{ times}} = |G|$$

Counting Orbits Example

(2020, Question 2 iii)

Let G be a finite group acting transitively upon a finite set with more than one element. Show that there is an element $g \in G$ without fixed points.

Hence,

$$\frac{1}{|G|} \sum_{g \in G} |S^g| > 1,$$

which is a contradiction. It follows that at least one element $g \in G$ must not have fixed points.

Centraliser and Class Equation

Working with the Center

We can say for group G , the center denoted $Z(G)$ is the set of elements that commute with every element. That is,

$$Z(G) = \{g \in G : hgh^{-1} = g \ \forall h \in G\}.$$

Equivalently, this is the set of elements with trivial orbits under the conjugation action (i.e. their conjugacy class is trivial, and their centraliser is all of G).

Theorem (Class Equation)

Let $X_1, \dots, X_r$ be the conjugacy classes of G with more than one element. Then

$$|G| = |Z(G)| + \sum_{i=1}^r [G : C_G(x_i)].$$

p -Groups

Definition (p -group)

Let p be a prime. A group G is called a p -group if $|G| = p^n$ for some $n \in \mathbb{N}$.

Results on p -groups

Let G be a p -group of order p^n , $n > 0$.

- (a) $Z(G)$ is a nontrivial subgroup.
- (b) For every $1 \leq m \leq n$, there exists a normal subgroup $N \leq G$ of order p^m .

Example p -Group Question

Example

Suppose that G has order p^3 and is non-abelian. Prove that $G/Z(G) \cong \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$.

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We first establish two claims.

Claim 1: If G has order p^n , then $Z(G)$ is nontrivial.

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Suppose that G has order p^3 and is non-abelian. Prove that $G/Z(G) \cong \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$.

We first establish two claims.

Claim 1: If G has order p^n , then $Z(G)$ is nontrivial.

This follows from the class equation, since each term $[G : C_G(x)]$ is either 1 (if $x \in Z(G)$) or a multiple of p . Taking the equation modulo p shows that $|Z(G)| \equiv |G| \equiv 0 \pmod{p}$. Since $Z(G)$ contains the identity, it must be a nontrivial multiple of p .

Example p -Group Question

Example

Suppose that G has order p^3 and is non-abelian. Prove that $G/Z(G) \cong \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$.

Using Claim 1, we prove:

Claim 2: A group of order p^2 is abelian.

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Using Claim 1, we prove:

Claim 2: A group of order p^2 is abelian.

From Claim 1, the centre has order p or p^2 . It suffices to show that if $|Z(G)| = p$, we obtain a contradiction.

Take $x \notin Z(G)$. Then its centraliser $C_G(x) = \{g \in G : gxg^{-1} = x\}$ contains x and all elements of $Z(G)$, so has size at least $p + 1$. Since $|C_G(x)|$ divides p^2 , it must be p^2 , so $C_G(x) = G$. Hence $x \in Z(G)$, a contradiction.

Example p -Group Question

Example

Suppose that G has order p^3 and is non-abelian. Prove that $G/Z(G) \cong \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$.

Alternatively, if $|Z(G)| = p$, then $|G/Z(G)| = p^2$. If $G/Z(G)$ is cyclic, then G is abelian.

Indeed, if $G/Z(G) = \langle gZ(G) \rangle$, then any $x, y \in G$ can be written as $x = g^i z_1$, $y = g^j z_2$ with $z_1, z_2 \in Z(G)$. Since elements of the centre commute with everything,

$$xy = g^{i+j} z_1 z_2 = yx.$$

Example p -Group Question

Example

Suppose that G has order p^3 and is non-abelian. Prove that $G/Z(G) \cong \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$.

Returning to the original problem, the centre must have order p , p^2 , or p^3 . By a similar argument to Claim 2, if $|Z(G)| = p^2$ or p^3 , then G is abelian, a contradiction. Hence $|Z(G)| = p$.

Thus $|G/Z(G)| = p^2$, and since every group of order p^2 is abelian, by the classification theorem we have

$$G/Z(G) \cong \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z} \quad \text{or} \quad \mathbb{Z}/p^2\mathbb{Z}.$$

The latter is cyclic, which would imply G is abelian, a contradiction. Hence the result follows.

Sylow Theorems

Sylow First Theorem

There exists a subgroup $P \leq G$ of order p^n (called a Sylow p -subgroup).

Sylow Second Theorem

If P_1, P_2 are Sylow p -subgroups of G , then they are conjugate.

Sylow Third Theorem

Let n_p be the number of Sylow p -subgroups. Then

$$n_p \mid m \quad \text{where } |G| = p^n m \text{ and } \gcd(p, m) = \gcd(p^n, m) = 1,$$

and

$$n_p \equiv 1 \pmod{p}.$$

Useful Sylow Lemmas

Lemma

If a finite p -group H acts on a set S , then $|S^H| \equiv |S| \pmod{p}$.

Lemma

If $Q \leq G$ is a p -subgroup and $P \leq G$ is a Sylow p -subgroup, then there exists $g \in G$ such that $gQg^{-1} \leq P$.

Result

If G is finite, p is prime, and $p^k \mid |G|$, then G has a subgroup of order p^k .

Sylow Theorems Example I

Original Example

If n_p is the number of Sylow p -subgroups of a group G , then there exists a subgroup H such that $[G : H] = n_p$.

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To motivate this, we want a way to obtain a result about the size of a quotient group. Lagrange's theorem naturally comes to mind, together with the First Isomorphism Theorem. However, since we are dealing with Sylow subgroups, the Orbit–Stabiliser Theorem is more natural.

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By the Second Sylow Theorem, we can view n_p as the size of an orbit for the action $G \curvearrowright X$, where X is the set of Sylow p -subgroups of G .

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If n_p is the number of Sylow p -subgroups of a group G , then there exists a subgroup H such that $[G : H] = n_p$.

Specifically, let P be any Sylow p -subgroup. First, observe that for any $g \in G$, the subgroup gPg^{-1} has the same order, and the Second Sylow Theorem tells us that every Sylow p -subgroup arises in this way. Hence, the size of the orbit $G \cdot P$ is n_p .

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Then, by the Orbit–Stabiliser Theorem,

$$n_p = |G \cdot P| = \frac{|G|}{|\text{Stab}_G(P)|} = [G : N_G(P)],$$

noting that $\text{Stab}_G(P) = N_G(P)$ as observed earlier. Hence, we are done.

Sylow Theorems Example II

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Let $H \leq G$ and let P be a Sylow p -subgroup of H . If $N_G(P) \leq H$, prove that P is a Sylow p -subgroup of G .

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We will use the action of $P \curvearrowright G/P$. Suppose that P is not a Sylow p -subgroup of G . Then we would have $|G/P| \equiv 0 \pmod{p}$.

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Since P is a p -group, we can use the fact that

$$|(G/P)^P| \equiv |G/P| \equiv 0 \pmod{p}.$$

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Further, it can be verified that $(G/P)^P = N_G(P)/P$. Hence $|N_G(P)/P| \equiv 0 \pmod{p}$, so it is a multiple of p . This can't be a zero multiple since $N_G(P)/P$ is non-empty (contains P).

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Since $N_G(P) \leq H$, we have $|N_G(P)| \mid |H|$ by Lagrange's theorem. Thus there must be an extra factor of p in $|H|$ beyond $|P|$, contradicting the assumption that P is a Sylow p -subgroup of H .

Hence, P is a Sylow p -subgroup of G .

Sylow Example

Sample Midterm Question 5

Suppose that G is a p -group of size p^n and that $P \leq G$ is a subgroup of size p^{n-1} . Then prove P is normal in G .

Consider the action $P \curvearrowright G/P$ by multiplication. Then for any $s \in G/P$, we would have by the orbit-stabiliser theorem that $|P \cdot s|$ divides $|P| = p^{n-1}$, so is a multiple of p .

Further observe that since the orbits partition $S = G/P$, which has size p , we have

$$\sum_{s \text{ representative of orbit}} |P \cdot s| = p.$$

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Further observe that since the orbits partition $S = G/P$, which has size p , we have

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But since the identity element P has a singleton orbit (just P), we must have that each orbit is trivial. That is, for all $g \in G$ and $h \in P$,

$$h \cdot gP = gP.$$

Hence $g^{-1}hgP = P$, so $g^{-1}hg \in P$, meaning that P is normal.

Questions for Break (I)

Quotient subgroup structure

Suppose that $H \trianglelefteq G$ such that $|G| = 4|H|$, and $K \leq G$. Prove that if

$$K/(K \cap H)$$

is non-trivial then it is isomorphic to either $\mathbb{Z}_2$, $\mathbb{Z}_2 \times \mathbb{Z}_2$, or $\mathbb{Z}_4$.

Automorphisms are inner on H

Recall that if $H, N \leq G$ then

$$HN = \{hn : h \in H, n \in N\}.$$

Suppose that $K \leq H \trianglelefteq G$ such that every automorphism of H is inner. Prove that

$$G = HN_G(K).$$

Questions for Break (II)

Heisenberg group quotient

Let $a, b, c \in \mathbb{Z}/3\mathbb{Z}$ and define

$$H = \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}.$$

It can be shown that H is a group under multiplication. Find a familiar group isomorphic to $H/Z(H)$.

Questions for Break (III)

Extension problems (home exercise)

Prove or disprove the following statements:

- (a) If $N_1 \trianglelefteq G_1$ and $N_2 \trianglelefteq G_2$ with $N_1 \cong N_2$ and $G_1/N_1 \cong G_2/N_2$, then $G_1 \cong G_2$.
- (b) If $N_1 \trianglelefteq G_1$ and $N_2 \trianglelefteq G_2$ with $G_1 \cong G_2$ and $N_1 \cong N_2$, then $G_1/N_1 \cong G_2/N_2$.
- (c) If $N_1 \trianglelefteq G_1$ and $N_2 \trianglelefteq G_2$ with $G_1 \cong G_2$ and $G_1/N_1 \cong G_2/N_2$, then $N_1 \cong N_2$.